# Kubernetes and Istio

Comprehensive Technical Documentation

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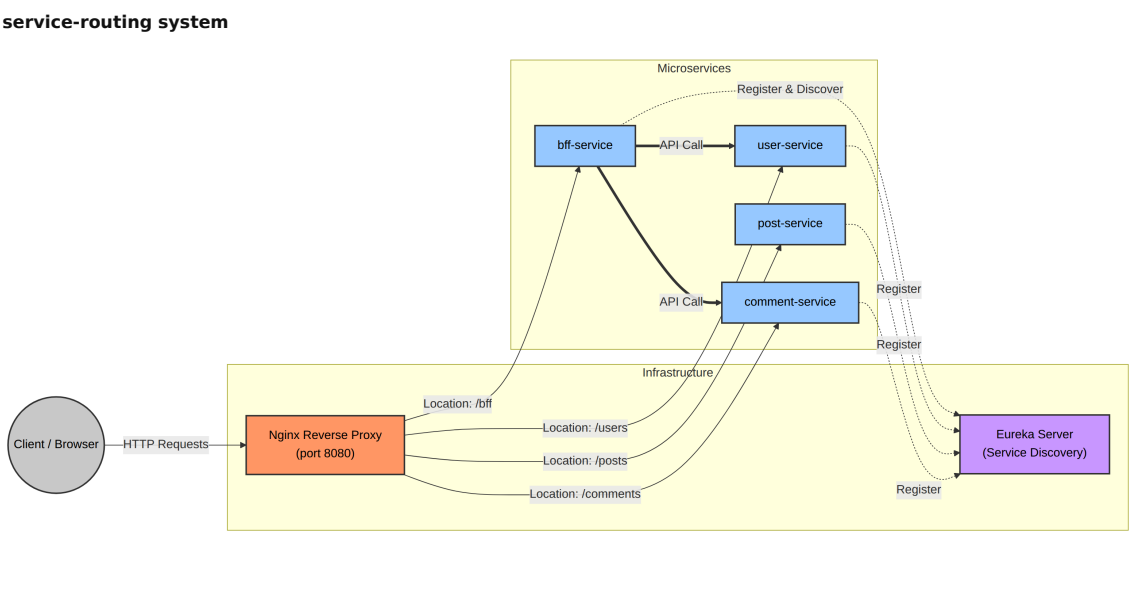
# 9 - First Istio architecture

In this lesson, you will learn how to use Istio instead of Eureka and Nginx in the `service-routing` example. Indeed, Kubernetes CoreDNS can seamlessly replace Eureka for service discovery, while the Istio Gateway can replace the Nginx reverse proxy. In addition, Istio provides secure communication between pods (**mTLS**) and powerful observability out of the box.

**service-routing system**


Microservices
Register & Discover
bff-service
API Call
user-service
post-service
API Call
comment-service
Register
Register
Infrastructure
Location: /bff
Client / Browser
HTTP Requests
Nginx Reverse Proxy
(port 8080)
Location: /users
Location: /posts
Location: /comments
Eureka Server
(Service Discovery)
Register

# Migrating to Kubernetes & Istio

To adapt the existing system to fit an Istio-enabled Kubernetes cluster, we need to perform the following actions:

1. **Remove Nginx Reverse Proxy**: Remove the `nginx` module from your parent build files.
2. **Remove Eureka Service Discovery Server**: Delete the `eureka-service` module completely.
3. **Remove Eureka Dependencies from the microservices**
   1. Remove the `Eureka` and `Spring Cloud` dependencies from `pom.xml` .
   2. Remove Eureka-related configurations from `application.yml` .
   3. Remove any `@LoadBalanced` annotations in the Java codebase (**Kubernetes and Istio will now handle load balancing natively**)
   4. Modify your routing URIs to match the new environment. Thanks to CoreDNS, each microservice can be directly reached at `http://<microservice-name>:<port>` .
4. **Compile the Applications**: Build your Java services using `mvn clean package -Dmaven.test.skip=true` .
5. **Containerize**: Build the Docker images and push them to your container registry.

# Define the infrastructure

Now that the container images are ready, we will define the Kubernetes YAML manifests. Each microservice is deployed using a `Deployment` and exposed internally via a `Service` . Here is an example for the `bff-service` :

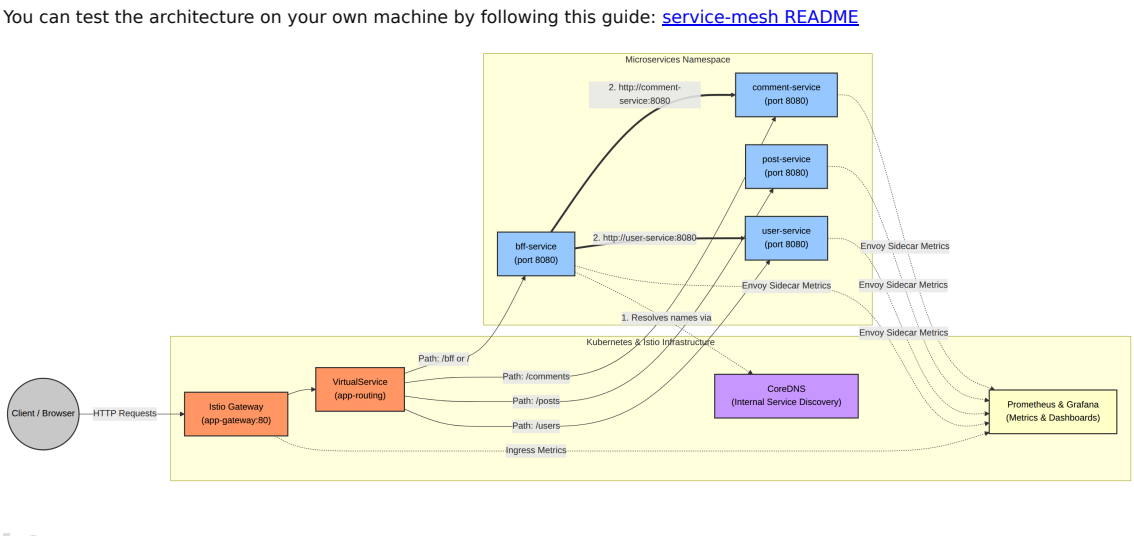
```yaml
apiVersion: v1
kind: Service
metadata:
  name: bff-service
  labels:
    app: bff-service
spec:
  ports:
    - port: 8080
      targetPort: 8080
      name: http
  selector:
    app: bff-service
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: bff-service
spec:
  replicas: 1
  selector:
    matchLabels:
      app: bff-service
  template:
    metadata:
      labels:
        app: bff-service
    spec:
      containers:
        - name: bff-service
          image: mattev02/bff-service:latest
          ports:
            - containerPort: 8080
          env:
            - name: SERVER_PORT
              value: "8080"
          readinessProbe:
            httpGet:
              path: /actuator/health
              port: 8080
            initialDelaySeconds: 10
            periodSeconds: 10
          livenessProbe:
            httpGet:
              path: /actuator/health
              port: 8080
            initialDelaySeconds: 20
            periodSeconds: 10
```

To route external traffic into the cluster, we use Istio Gateway API. A `Gateway` handles the entry point (replacing standard Ingress objects), and an `VirtualService` directs the traffic to the corresponding microservices based on the requested path.

```yaml
apiVersion: networking.istio.io/v1
kind: Gateway
metadata:
   name: app-gateway
spec:
   selector:
      istio: ingressgateway
   servers:
      - port:
           number: 80
           name: http
           protocol: HTTP
        hosts:
           - "*"
---
apiVersion: networking.istio.io/v1
kind: VirtualService
metadata:
   name: app-routing
spec:
   hosts:
      - "*"
   gateways:
      - app-gateway
   http:
      # 1. The BFF Route
      - match:
           - uri:
                prefix: /bff
        route:
           - destination:
                host: bff-service
                port:
                   number: 8080

      # 2. The Users Route
      - match:
           - uri:
                prefix: /users
        route:
           - destination:
                host: user-service
                port:
                   number: 8080

      # 3. The Posts Route
      - match:
           - uri:
                prefix: /posts
        route:
           - destination:
                host: post-service
                port:
                   number: 8080

      # 4. The Comments Route
      - match:
           - uri:
                prefix: /comments
        route:
           - destination:
                host: comment-service
                port:
                   number: 8080

      # 5. Default catch-all route fallback
      - route:
           - destination:
                host: bff-service
                port:
                   number: 8080
```

# The new architecture

The final system architecture is represented below.
You can test the architecture on your own machine by following this guide: service-mesh README


Microservices Namespace
2. http://comment-service:8080
comment-service
(port 8080)
post-service
(port 8080)
bff-service
(port 8080)
2. http://user-service:8080
user-service
(port 8080)
Envoy Sidecar Metrics
Envoy Sidecar Metrics
Envoy Sidecar Metrics
Envoy Sidecar Metrics
1. Resolves names via
Kubernetes & Istio Infrastructure
Path: /bff or /
Client / Browser
HTTP Requests
Istio Gateway
(app-gateway:80)
VirtualService
(app-routing)
Path: /comments
Path: /posts
Path: /users
CoreDNS
(Internal Service Discovery)
Prometheus & Grafana
(Metrics & Dashboards)
Ingress Metrics

> ❓ ***What are the advantages of this system compared to the initial one?***
>
> - ***Reduced codebase****: Removes boilerplate code and eliminates the need for maintaining a standalone Eureka server.*
> - ***Language Agnosticism****: While Eureka ties you heavily to the Java ecosystem, Istio works natively with applications written in any language.*
> - ***Offloading Resilience to Infrastructure****: In a traditional Spring Boot app, you must configure libraries like Resilience4j and write fallback logic in Java. Istio manages retries, timeouts, and circuit breaking at the platform level.*
> - ***Simplified Observability****: Metrics, tracing, and logging are captured automatically by the Envoy sidecars.*
> - ***Encrypted Communications****: Istio transparently enforces mTLS between pods, securing internal traffic.*
> - ***Zero-Trust Security****: Istio allows you to define strict* `AuthorizationPolicies` *. For example, you can write a rule stating: "Only bff-service is allowed to call user-service. Reject traffic from any other pod."*